

OPNsense: Setting Up the Basic Firewall

Security concerns prompt the securing of networks against hacking and intrusion.

OPNsense is an open source, FreeBSD based firewall and routing platform. It includes most of the features available in expensive commercial firewalls, and often even more. It combines the rich feature set of commercial offerings with the benefits of open and verifiable sources.



Network monitoring, traffic investigation, packet analysis and developing robust firewalls are the key aspects of digital forensics because of increasing threats and vulnerabilities. Efforts are being made to develop and deploy fully secured network infrastructure, but the escalating incidents of cyber terrorism and cracking attempts need constant alertness.

While a number of software as well as hardware components with the latest updates are released frequently, yet, there is a need to be conscious of every aspect of security. Proprietary firewalls and security devices are used in the corporate world as well as by government organisations, but these are very costly in terms of annual maintenance as well as the initial one-time investment.

In classical methods of implementation, network administrators make use of packet capturing tools and intrusion detection systems for deep analysis of the packets flowing in the network. These network administration and forensic activities are implemented using PCAP and related IDS tools, which include free and open source software as well as commercial proprietary products.

The key WinPcap/libpcap software suites working with PCAP at the base level include:

WinDump	Snort	ngrep
Wireshark	Nmap	LOphtCrack
Bro IDS	XLink Kai	iftop
EtherApe	URL Snooper	Kismet
Tcpdump	Bit-Twist	Pirni
Suricata	NetSim	Firesheep
WhatPulse	Xplico	Scapy

There are a number of intrusion detection systems (IDS) used in corporate as well as research organisations to enforce and implement threat analysis and digital security. Some of these IDS products are:

AIDE	Samhain	OpenDLP
OSSEC	Security Onion	OpenWIPS-NG
ACARIM-ng	Fail2ban	OSSEC HIDS
Bro NIDS	Prelude Hybrid IDS	Samhain
Snort	Suricata	

OPNsense: A free and open source firewall

URL: opnsense.org

OPNsense is a powerful and user-friendly firewall